

Reducing Patient Waiting Times Through a Digital Queue Management System in Clinics and Hospitals

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I. Introduction

Waiting for healthcare services is a common experience for patients visiting clinics and hospitals. Long waiting times can occur at different stages of the healthcare process, including registration, consultation, referrals, laboratory services, and other procedures. Aside from being inconvenient, prolonged waiting can influence how patients perceive the quality and efficiency of healthcare services. Waiting time is considered an important indicator of healthcare service delivery and patient satisfaction (Bontia & Babsa-ay, 2024).

The problem is particularly relevant in outpatient and specialty services where the number of patients, available healthcare personnel, scheduling arrangements, and service capacity may vary throughout the day. A study conducted at the Philippine General Hospital found that waiting times for specialty referrals varied according to factors such as the urgency of the case, type of specialty clinic, and purpose of referral (Bontia & Babsa-ay, 2024). These differences demonstrate how patient flow and the organization of healthcare services can contribute to delays.

Technology provides an opportunity to improve the way patients are managed while waiting. Digital registration, appointment systems, queue monitoring, and electronic notifications can provide patients with clearer information about their place in the queue while helping healthcare facilities organize patient flow. Studies examining electronic queue and online registration systems have reported improvements in waiting time, patient satisfaction, and administrative efficiency (Choosri et al., 2022; Sharma et al., 2022).

This proposal therefore focuses on the development of a Digital Queue Management System (DQMS) designed to organize patient queues, provide real-time queue information, and notify patients when their turn is approaching.

II. Problem Description

Long waiting times in clinics and hospitals are a significant healthcare service problem because patients may spend a considerable amount of time waiting before receiving the service they need. Waiting is also more than simply an inconvenience because uncertainty about how long a patient must wait can negatively affect satisfaction. Zhang et al. (2023) explained that long waiting times are associated with lower patient satisfaction and can affect patients' confidence in healthcare facilities.

Several factors can contribute to prolonged waiting times. These include high patient demand, limited healthcare resources, differences in the urgency and type of service required, scheduling issues, and the movement of patients between different stages of care. In a Philippine tertiary government hospital, Bontia and Babsa-ay (2024) found that referral waiting times differed according to urgency, specialty clinic, purpose of referral, and clinical characteristics. This shows that waiting times are influenced by both patient-related and healthcare-system factors.

Traditional queue systems can also contribute to the problem because patients may have to remain physically present in a waiting area without knowing exactly when they will be served. This can lead to crowded waiting areas and frustration, particularly when there is limited information about the current queue. Zhang et al. (2023) noted that uncertainty about waiting time can increase dissatisfaction, suggesting that providing patients with more accurate information about expected waiting times can improve their experience.

The problem is not limited to patient satisfaction. Long and inefficient queues can also increase administrative workload and create difficulties in managing patient flow. Healthcare workers may need to repeatedly answer questions about queue status or manually organize patients as they move through different service points. Therefore, addressing waiting time requires not only increasing service capacity but also improving how patients and queues are organized.

III. Proposed Solution

The proposed solution is the development of a Digital Queue Management System (DQMS) for clinics and hospitals. The system would use information technology to organize patient queues electronically and provide patients and healthcare personnel with real-time information about the status of the queue.

The system would begin when a patient registers for a service. Registration could be completed through a hospital kiosk, registration counter, or compatible mobile/web interface. After registration, the system would generate a digital queue number and assign the patient to the appropriate department or service.

Patients would then be able to monitor their queue position through a digital display or mobile interface. The system could show the current number being served, the patient's approximate position, and an estimated waiting time. When the patient's turn is approaching, the system could send an automated notification or display an alert instructing the patient to proceed to the appropriate service area.

For healthcare personnel, the system would provide a centralized queue dashboard. Staff could view the number of patients waiting, identify congested departments, update queue status, and manage patients according to the appropriate service. The system could also collect data on waiting times, allowing hospital administrators to identify recurring bottlenecks and improve scheduling and resource allocation.

The proposed DQMS would therefore focus on four primary functions:

1. **Digital Registration and Queue Assignment** – Patients receive an electronic queue number after registering for a service.
2. **Real-Time Queue Monitoring** – Patients can view their current queue status and estimated waiting time.
3. **Automated Notifications** – Patients receive alerts when their turn is approaching, reducing the need to remain continuously in the waiting area.
4. **Queue and Patient-Flow Dashboard** – Healthcare personnel can monitor queues, manage patient flow, and use collected data to identify delays and bottlenecks.

IV. Conclusion

Long waiting times in clinics and hospitals can negatively affect patient satisfaction and contribute to inefficiencies in healthcare service delivery. The problem can be influenced by patient volume, staffing and resource limitations, scheduling, the type and urgency of services, and the way patients move through the healthcare system.

Ultimately, the proposed system aims to transform the traditional waiting process from an uncertain and crowded experience into a more organized and transparent one. Although technology alone cannot solve every cause of healthcare delays, a well-designed digital queue system can help clinics and hospitals manage patient flow more effectively and provide patients with a better healthcare experience.

References:

- Bontia, A. A. I., & Babsa-ay, J. D. (2024). Factors affecting waiting time of patients referred to specialty clinics from a family medicine clinic in a tertiary government hospital: A retrospective chart review. *Acta Medica Philippina*, 58(13), 39–44. <https://doi.org/10.47895/amp.v58i13.8098>
- Zhang, H., Ma, W.-M., Zhu, J.-J., Wang, L., Guo, Z.-J., & Chen, X.-T. (2023). How to adjust the expected waiting time to improve patient's satisfaction? *BMC Health Services Research*, 23, 455. <https://doi.org/10.1186/s12913-023-09385-9>